

Biometric Authentication via Smartwatch

CS Summer Bootcamp Research Project

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SMU



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Paper Research

You Define Your Biometrics: Customized Biometric Authentication via Commercial Handheld Device

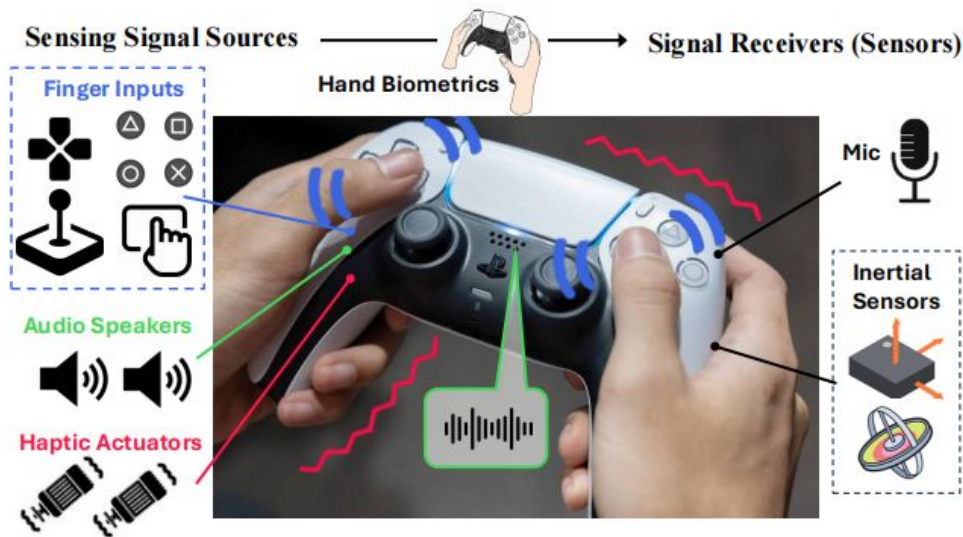
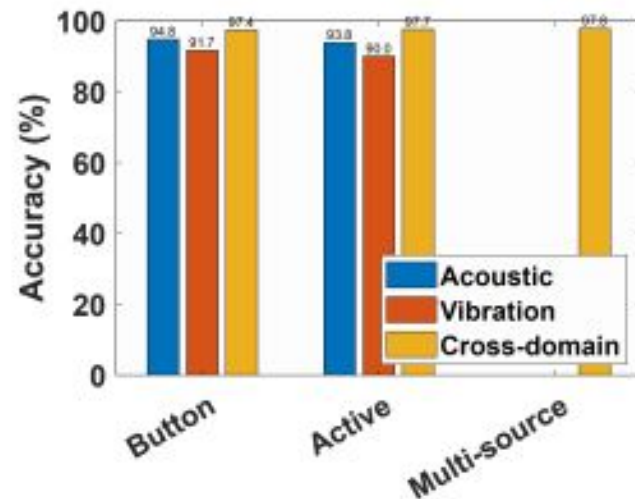


Fig. 1: Illustration of customized biometric authentication on any handheld device that has multi-modal sensing ability.



(a) Accuracy

Collect Data Platform



1. Require related permissions (media, health...)
2. Tap the button to start collecting; tap again to stop.
3. Save data to CSV per sensor.
4. Microphone audio is saved as .wav files.
5. Reset Data
6. Send to Phone

Development Process



Original package

Imported previous smartwatch sensor collector codebase.

Build & Runtime

Added Gradle project files, wrapper, and permission fixes.



Sensor expansion

Registered additional standard and specific sensors.



CSV Export

Save collected data to csv instead of database.



Audio Recording

Preserve the audio file as wav instead 3gp.



Audio player

Play swept-frequency signal (0 - 24KHz) once.



Reset stored data
& Send to Phone

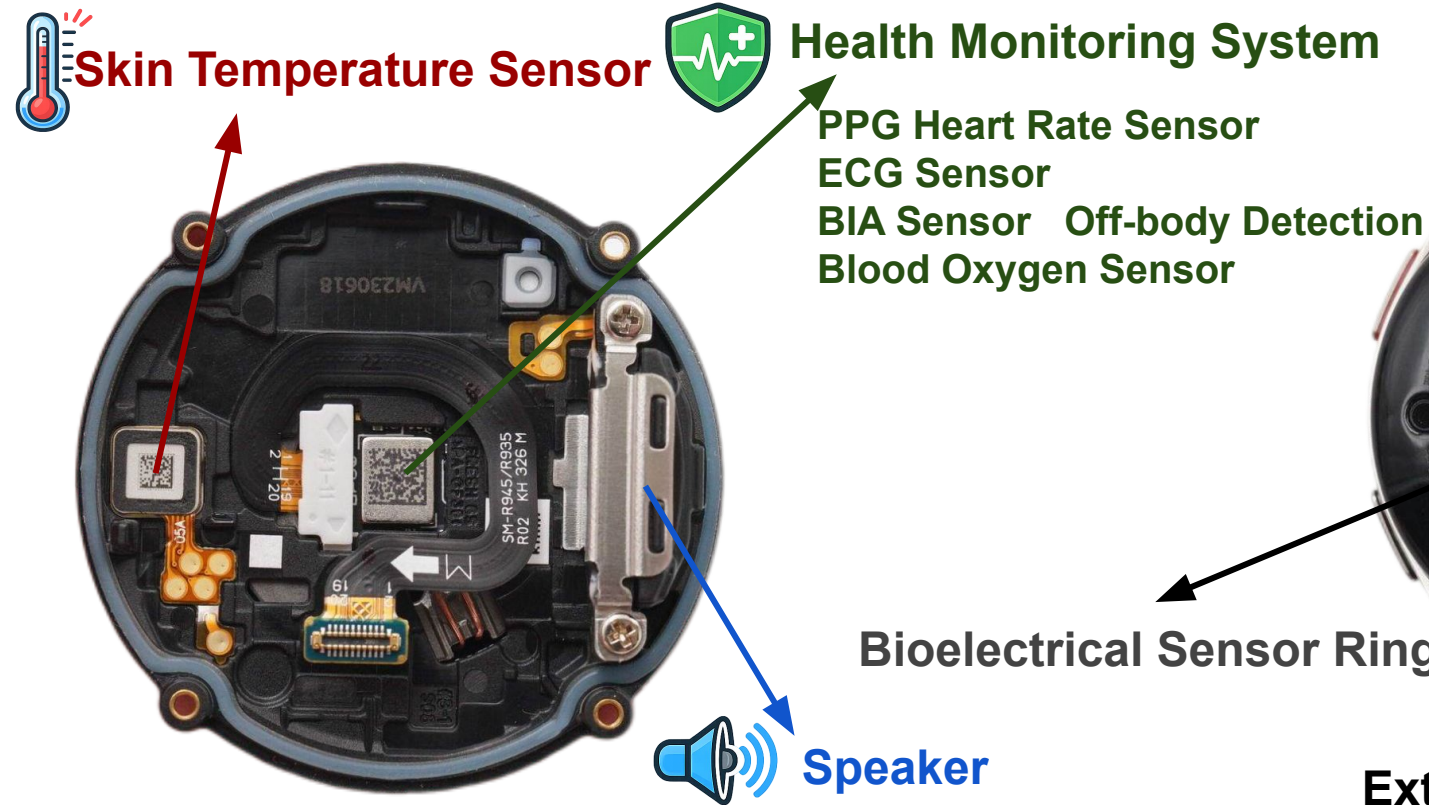
Add the button that can reset the watch local storage and “Send to Phone” button. Export files easily and separate test user files.



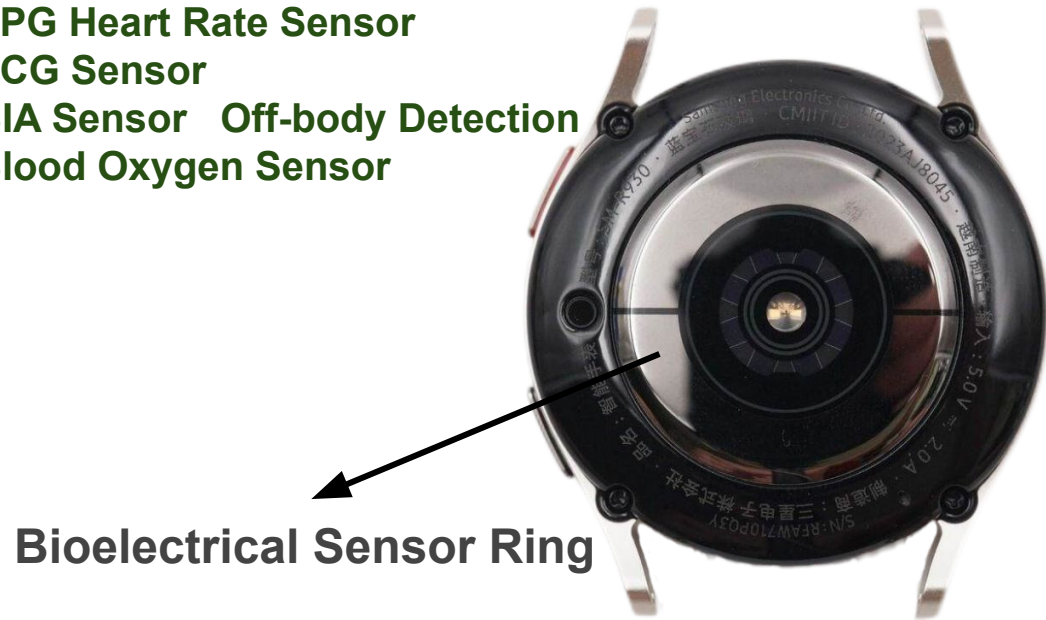
Sampling Rate

Improve sampling rate from 20Hz to 99Hz.

Sensor Coverage



Internal Sensor Layout

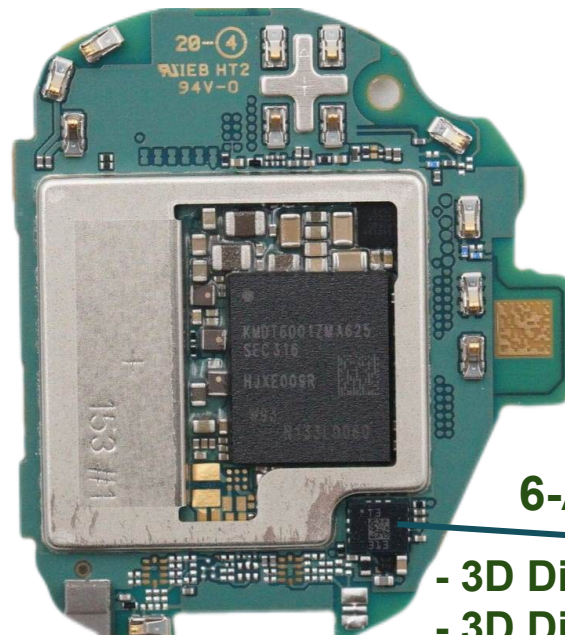


External Sensor Layout

Sensor Coverage



Microphone



6-Axis IMU Sensor

- 3D Digital Accelerometer
- 3D Digital Gyroscope

Main Board Assembly

Back Cover Assembly



Vibration Motor

Pressure Sensor

Source: 52audio. <https://www.52audio.com/archives/235478.html>

Samsung SDK

Samsung Health Data SDK Work with Phone Device

- BodyTemperature
- HeartRateType
- BloodOxygenType
- BloodPressureType etc

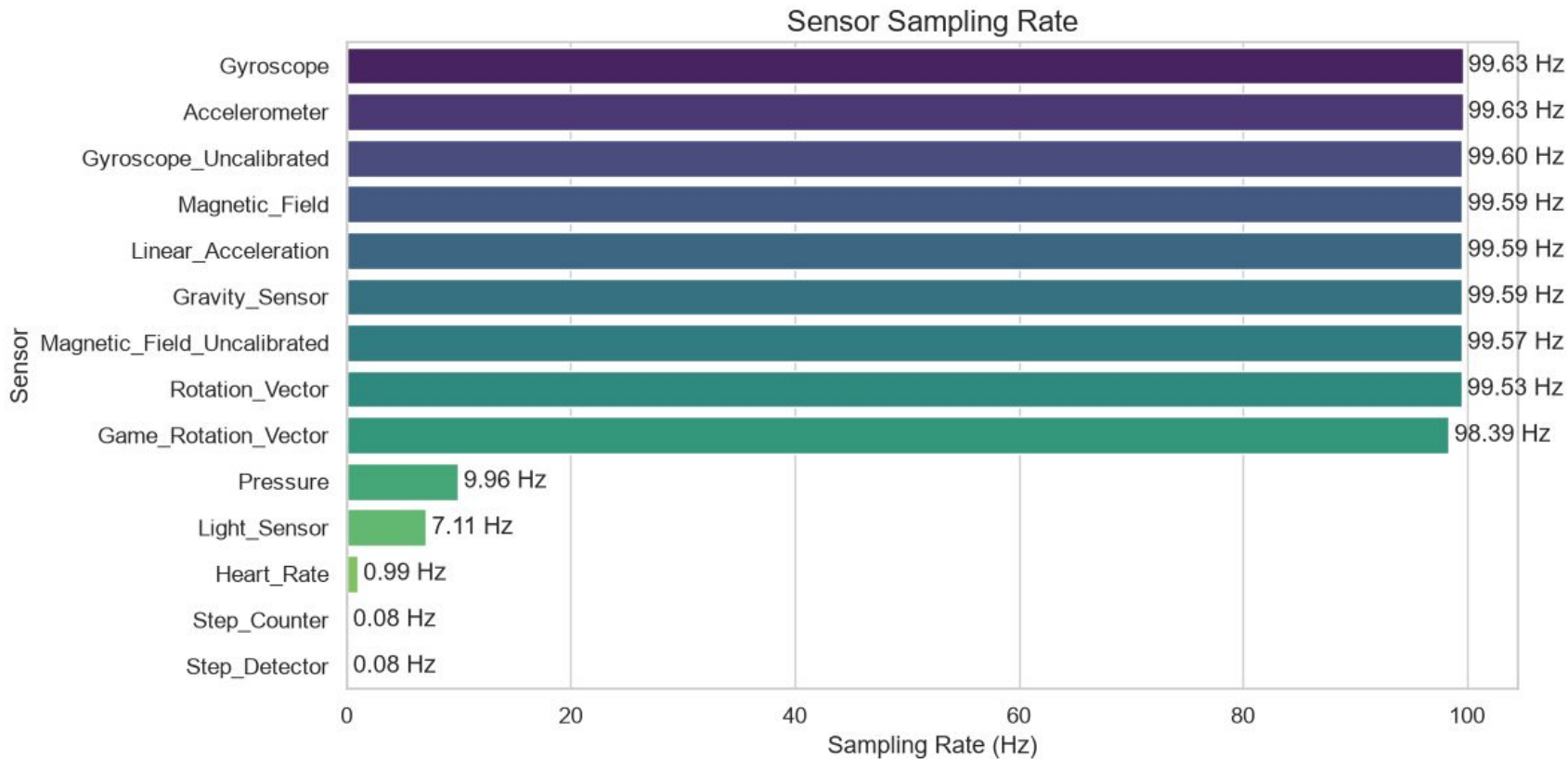
<https://developer.samsung.com/health/data/process.html>

Samsung Health Sensor SDK Need to request partnership Work with Watch Device

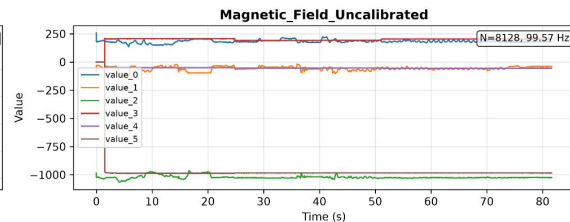
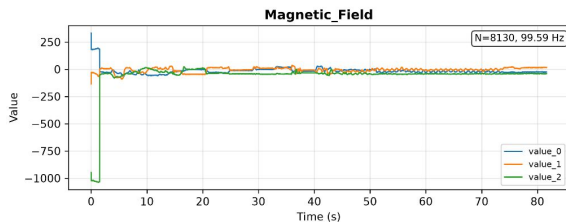
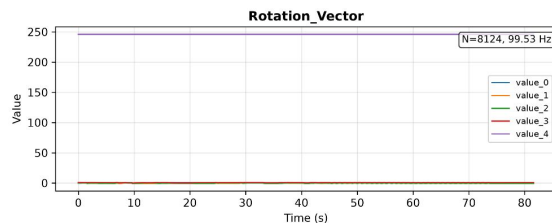
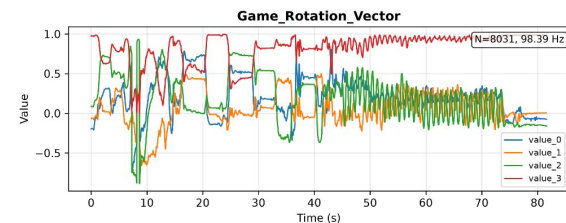
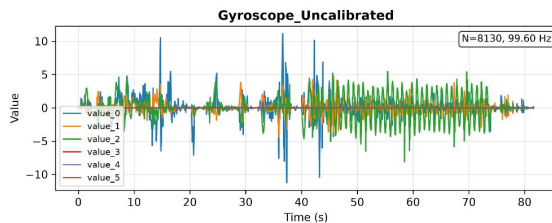
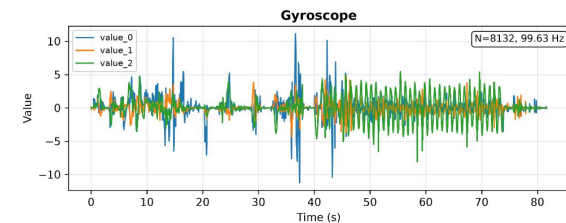
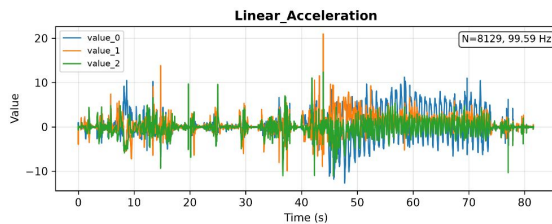
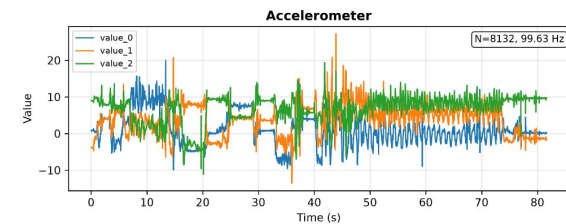
- Raw ECG (electrocardiogram)
- PPG (Photoplethysmogram)
- Heart rate, including inter-beat interval

<https://developer.samsung.com/health/sensor>

Sensor Sampling Rate



Motion Sensor Evaluation

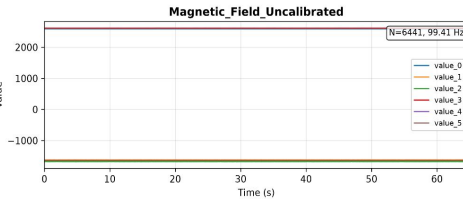
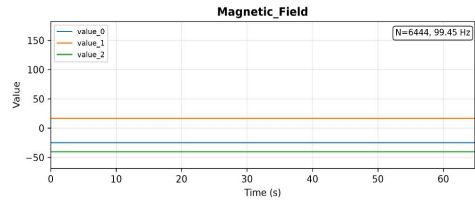
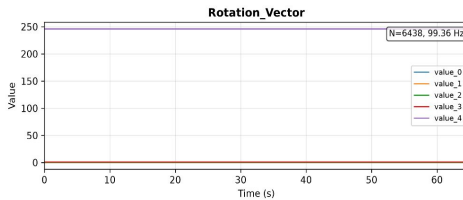
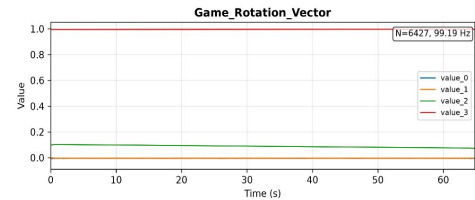
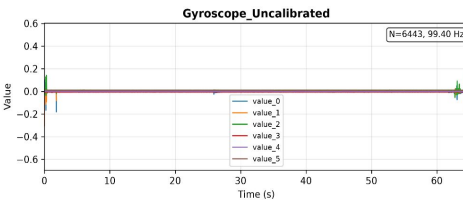
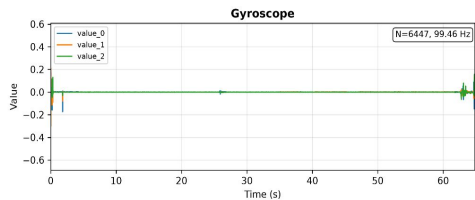
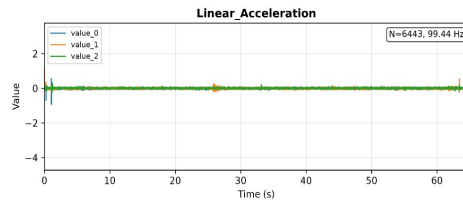
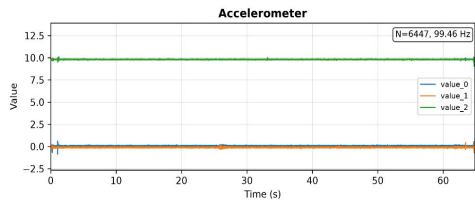


Sensors

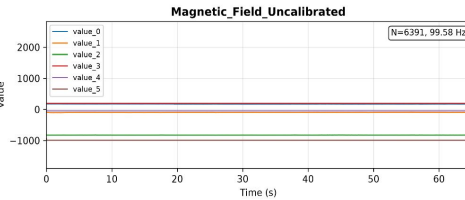
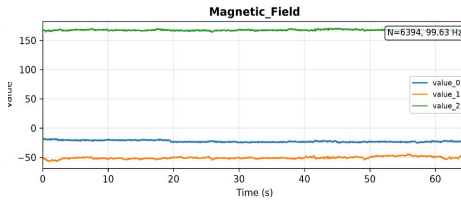
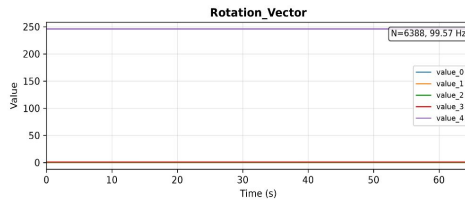
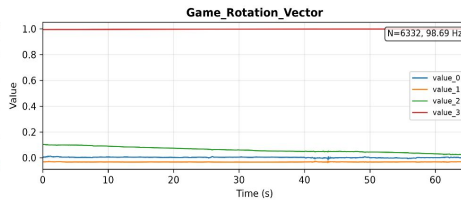
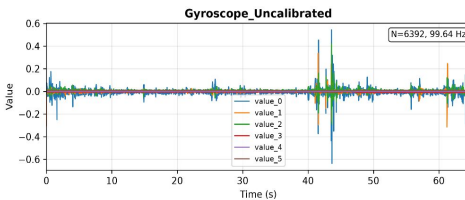
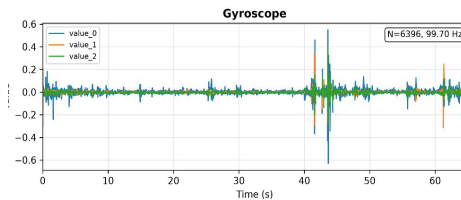
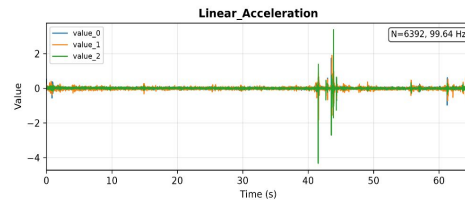
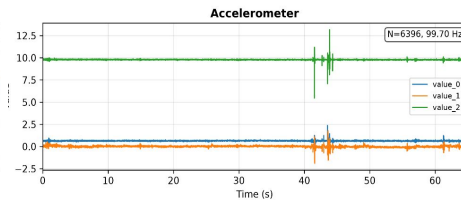
- Accelerometer
- Linear Acceleration
- Gyroscope
- Gyroscope Uncalibrated
- Rotation Vector
- Game Rotation Vector
- Magnetic Field
- Magnetic Field Uncalibrated

Motion Sensor Comparison

bodyoff - Motion Sensors

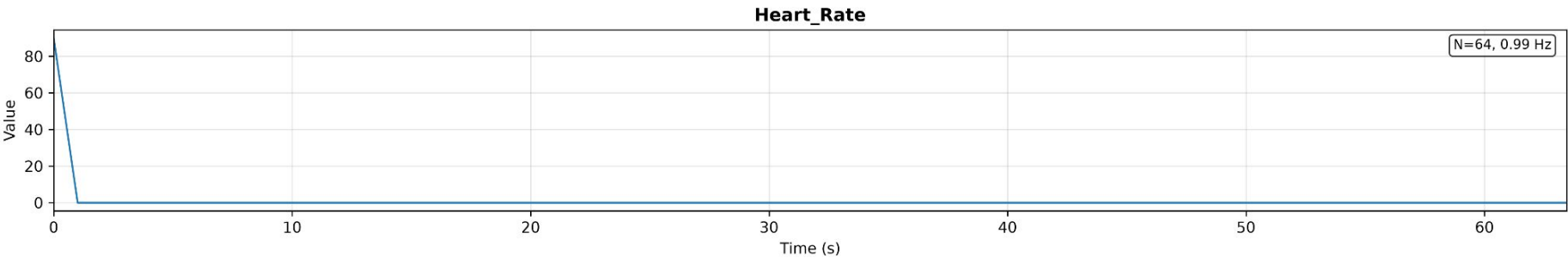


bodyon - Motion Sensors

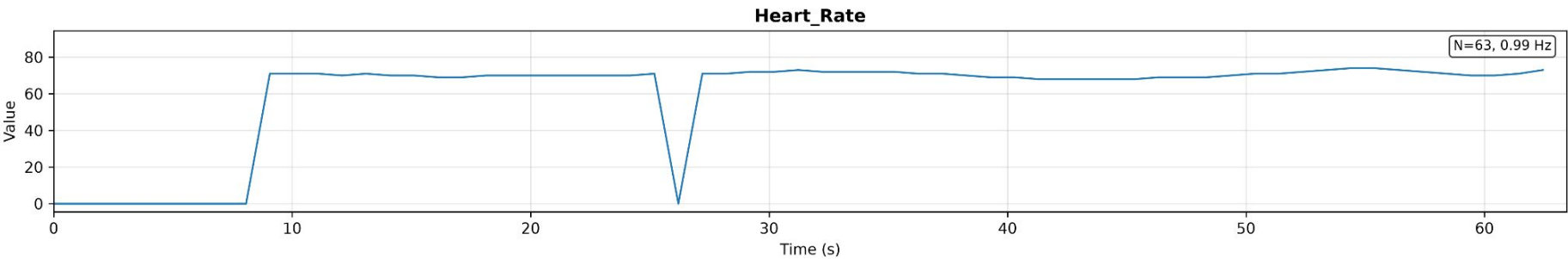


Body Sensor

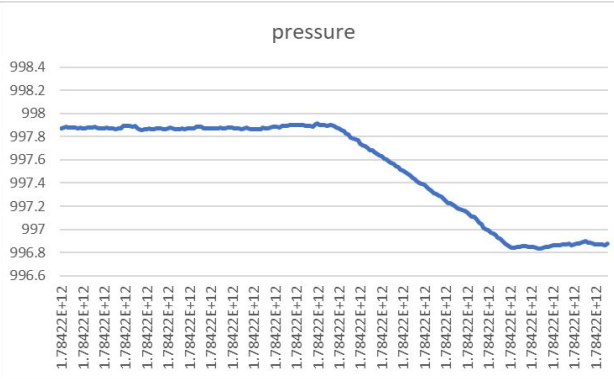
bodyoff - Body Sensors



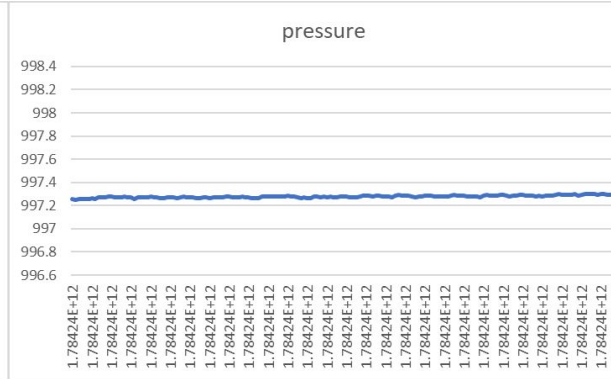
bodyon - Body Sensors



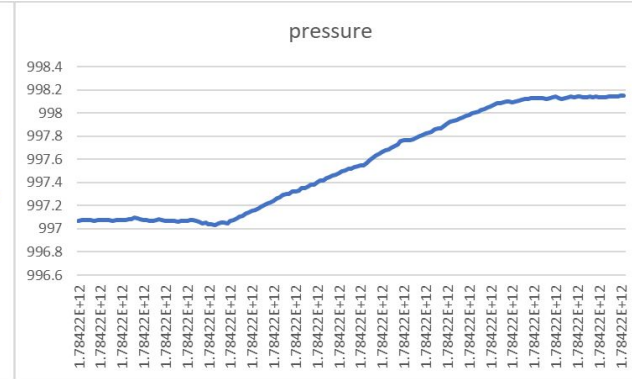
Pressure Sensor



Elevator Ascending



Stationary



Elevator Descending

Sensors

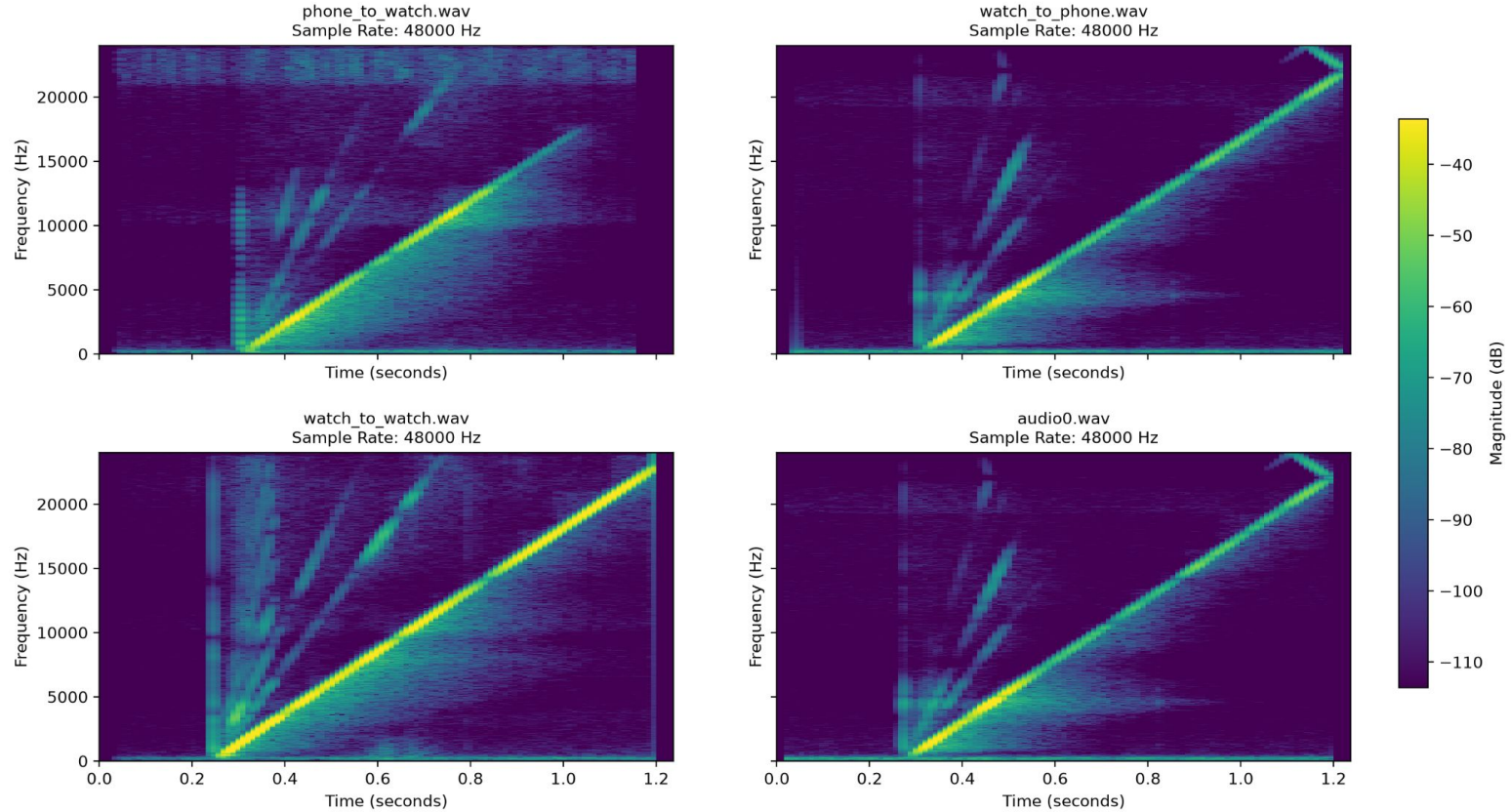
- Pressure

OBSERVATION

- Atmospheric pressure decreased during ascent.
- Pressure remained stable while stationary.
- Atmospheric pressure increased during descent.
- Height difference ≈ 8.3 m; pressure change ≈ 100 Pa, consistent with theoretical expectations.

Audio Sensor

Spectrogram Comparison



Data Collection Design

Collection Flow

1. Tap "Start" → the app immediately plays a 2-second, 0 – 48 kHz linear frequency sweep.
2. While the chirp plays, the app starts audio recording and 14-channel sensor logging.
3. The participant types "watch" on a computer while collecting.
4. Tap "Stop" → session ends, ~3–4 seconds total

Dataset Scale

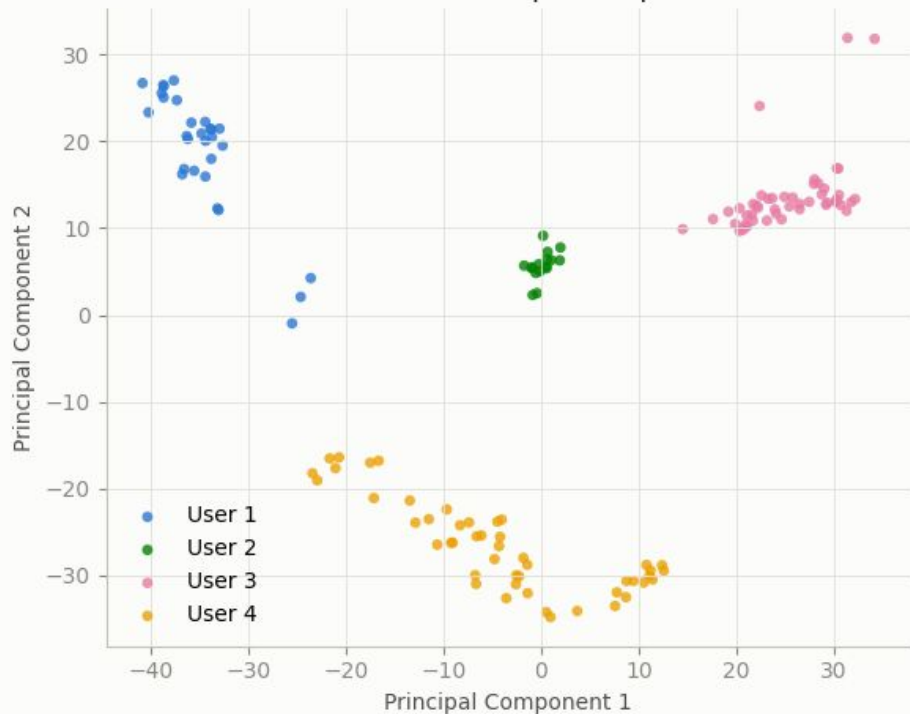
- 4 participants x 20 sessions each = 80 valid sessions
- Each session exports: 1 audio (.wav) file + 14 sensor CSV files
- Data pulled from the watch via adb pull, organized by participant, then anonymized to User 1–4 before training

Data Analysis and Classification

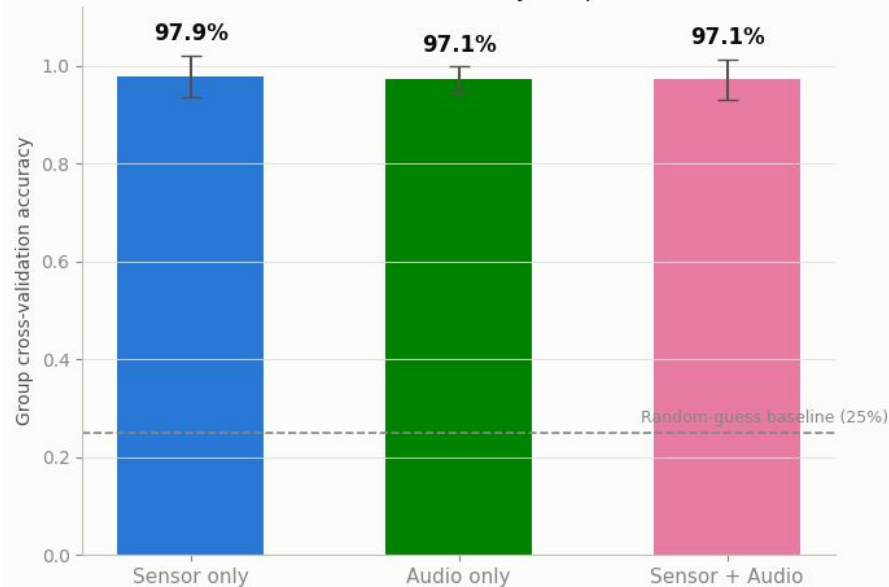
Data Preprocessing	Windowing	PCA	SVM
Read the sensor csv data and audio for every session. Resample each sensor to a common time grid. Align audio to sensor timestamps using the chirp signal.	Slice each session into fixed-length, overlapping time windows (e.g. 3s window, 1s step). Each window becomes one training sample, sensor data in the window is resampled to a fixed number of points.	Flatten each window's sensor data and combine it with the audio features (MFCC, RMS, zero-crossing rate, spectral shape). Standardize the combined feature vector and reduce dimensionality with PCA (retain 95% variance).	Train an RBF-kernel SVM on the PCA components to classify User A and User B. Evaluate with a session-grouped train/test split to avoid windows from the same session leaking across sets.

Data Analysis and Classification

PCA: First Two Principal Components



Cross Validation Accuracy Comparison



Limitation & Future Work

Limitations

- Small dataset: only 4 participants, 80 sessions total
- Fixed task: participants only typed the word "watch"
- Model restriction: only evaluated with SVM

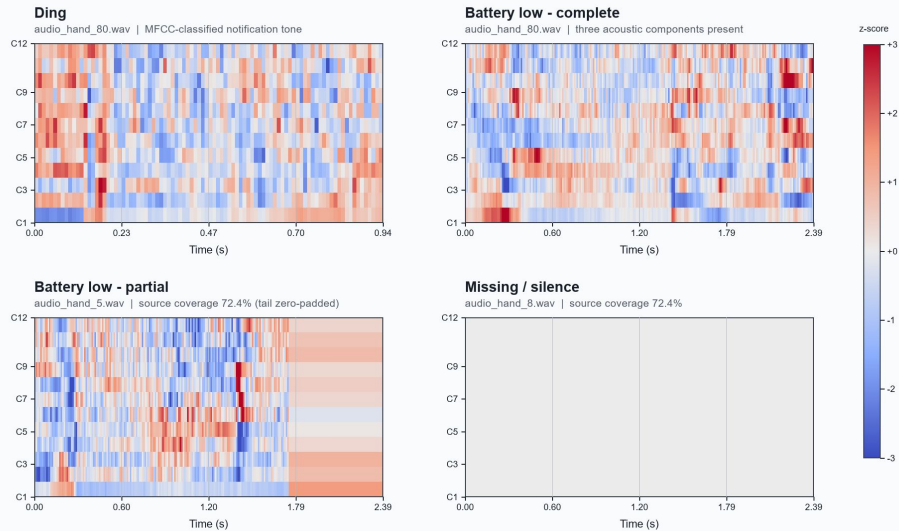
Future Work

- Scale up to more participants and sessions for stronger generalizability
- Collect free-form typing, multiple passwords, and varied gestures
- Compare SVM against deep learning architectures (e.g., BiLSTM + attention, as in the referenced paper)

Audio Echo Preprocessing

MFCC Heatmap Comparison

C1-C12 standardized over time | shared color scale

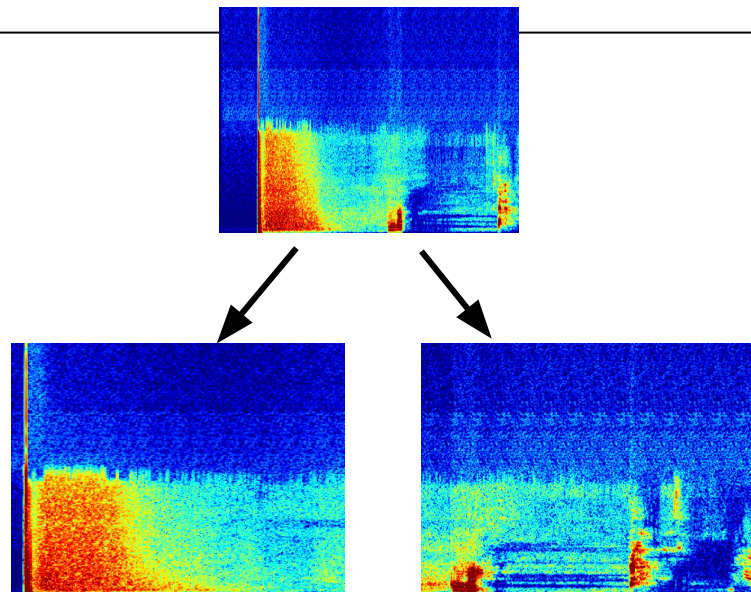


MFCC-based

- Audio event detection
- MFCC feature extraction
- Nearest-centroid classification
- High-confidence segment selection

Spectrogram-based

- Delay estimation & segmentation
- Spectrogram generation
- Cosine similarity classification
- Batch image generation



Thank You

Questions & Discussion

A wide-angle photograph of the Southern Methodist University (SMU) building, a large red brick structure with a prominent green dome and a classical portico with columns. The building is set against a blue sky with large, white, fluffy clouds. In the foreground, a green lawn is populated with several people sitting or walking. A small fountain is visible in the middle ground. The overall scene is bright and sunny.

SMU

SMU